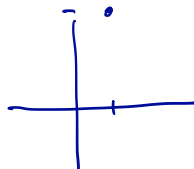


Algorithms and Analysis

What this course will / will not cover

Optimization:

$$\begin{array}{ll} \text{minimize :} & f(x) \\ \text{subject to :} & x \in \mathcal{X}. \end{array}$$

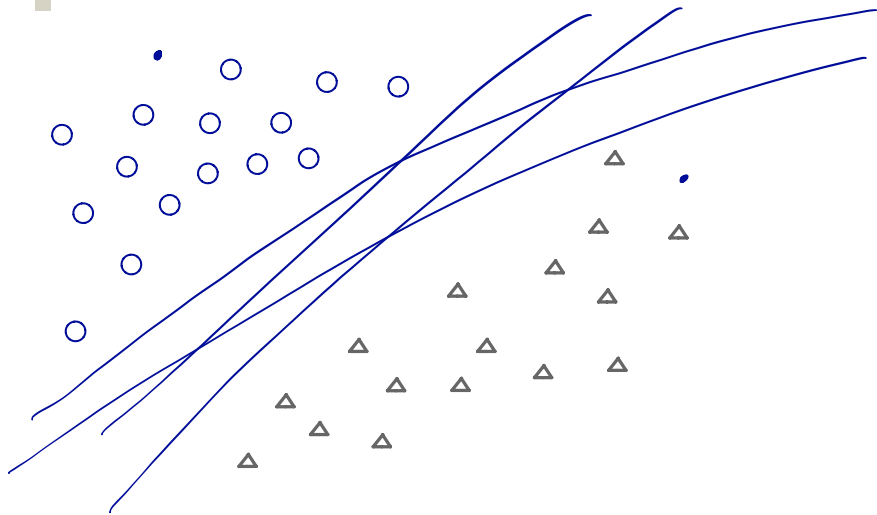


$$x \in \mathbb{R}^n : x = (x_1, x_2, \dots, x_n)$$

What can this model?

When can we solve it?

Max margin classification



Max margin classification

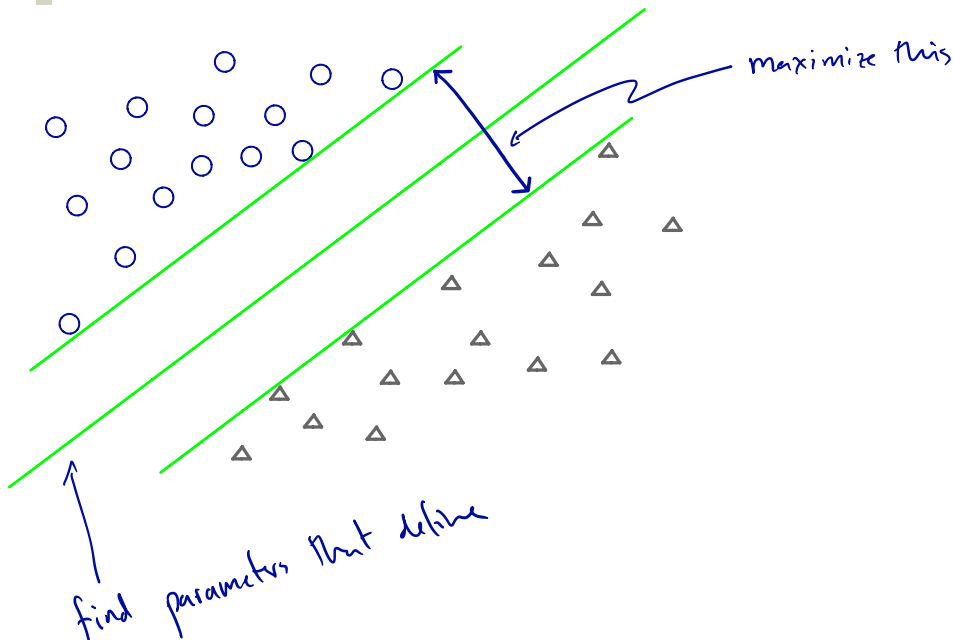


Image Denoising

Cost: $\text{distance from original} + \text{"how noisy image looks"}$



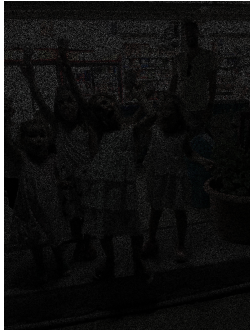
original



noisy

Image In-Painting

deleted 90% pixels



recover



original



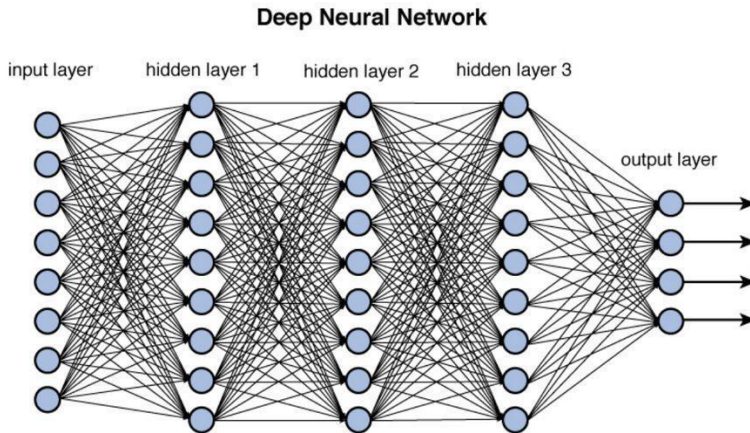
parameters : values of missing pixels
objective : "f(x)" : measure of how many sharp borders

Matrix Completion and Collaborative Filtering



F i l m s	Users						
	1		2			4	
		3		?	2		
	4		5	3			3
		2			3		
				5		2	

Neural Networks



$f(x)$ = classification error
 x = all the weights of the NN



And the list goes on and on...

Scheduling

Resource allocation

Combinatorial problems

Path planning

.....

.....

.....

Algorithms and Analysis

What this course will / will not cover

Optimization:

$$\begin{array}{ll} \text{minimize :} & f(x) \\ \text{subject to :} & x \in \mathcal{X}. \end{array}$$

What can this model?

When can we solve it? We need f and X to be convex

Algorithms and Analysis

What this course will / will not cover

Optimization:

$$\begin{array}{ll} \text{minimize :} & f(x) \\ \text{subject to :} & x \in \mathcal{X}. \end{array}$$

How do we model interesting problems using convex optimization? What are specialized solvers we can then call to solve these problems?

Algorithms and Analysis

What this course will / ~~will not~~ cover

Optimization:

$$\begin{array}{ll} \text{minimize :} & f(x) \\ \text{subject to :} & x \in \mathcal{X}. \end{array}$$

What are the basic algorithms for solving these convex problems? How do we develop new algorithms? How fast do they run, and what computing resources do they need?

Examples of problems this course will cover

M-Estimators —

- Least Squares Regression

- Ridge (L2-regularized) Regression

- Lasso (L1-regularized Regression)

- Logistic Regression

General convex functions for which we can compute derivatives.



In Summary: This class —

Fundamental (broad) problem classes

Algorithms tailored for these

Analysis of convergence rates

Implications for Machine Learning problems